

Geometry and Calculus for Computing

Module 1

Angles, triangles and trigonometry

Lesson 1.0 Introduction

 Course structure and navigation
Reading • 15 min How to learn effectively on this course
Reading • 15 min Introduction to the course
Video • 1 min Course Syllabus
Reading • 10 min

Lesson 1.1 Introduction to Angles, triangles and trigonometry

 Introduction to triangles and trigonometry
Video • 15 min The circle
Video • 8 min From the circle to the sine and cosine graphs
Video • 9 min Introducing the tangent
Video • 5 min Applying trigonometry to solve triangle problems
Dialogue • 30 min Introduction to triangles and trigonometry
Practice Assignment • 30 min

Lesson 1.2 Generic Triangles

 Applications of sine and cosine rules – examples
Video • 10 min Sine and cosine rules
Practice Assignment • 30 min

Lesson 1.3 Trigonometric Functions

 Unit circumference and definition of trigonometric functions for every angle
Video • 9 min Plotting tan
Video • 2 min Unit circumference and definition of trigonometric functions for every angle
Practice Assignment • 30 min Trigonometric functions, plots and properties
Video • 19 min Trigonometric functions, plots and properties
Practice Assignment • 30 min

Lesson 1.4 Summary and Assessments

 Check your understanding: End of module 1
Graded Assignment • 20 min Summary
Reading • 10 min

Module 2

Graph sketching and kinematics

Module 3

Exponential and logarithmic functions

Module 4

Limits and differentiation

Course structure and navigation

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Welcome to the Geometry and Calculus for Computing course and thank you for joining us. This course has been developed by **Goldsmiths, University of London** in collaboration with [University of London Worldwide](#), and we're excited to share it with a global community of learners. Whether you're here to develop your knowledge, develop new skills or explore a personal interest, we hope the journey ahead will be both rewarding and enjoyable.

What this course offers

In this course, you'll be guided through the topic step by step using a mix of videos, readings, practical exercises and discussions.

To support your critical thinking, we've introduced the AI-powered Dialogue exercises. These are designed to help you engage more deeply with the material through simulated, guided conversations. Don't worry - we'll provide clear instructions when you reach them.

While there's no live tutor support, you won't be learning in isolation. Peer interaction and community learning are central to the course experience, and we encourage you to connect with others and learn together.

Before you get started, make sure to check the course page for a weekly overview of the topics you'll be covering, and visit the grades page to understand how your progress will be formally assessed.

Technical support and help

To get the most out of your learning experience, we recommend using the desktop version of Coursera, with the mobile app as a convenient companion when you're on the go.

Navigation issues: Visit the [Coursera Learner Help Centre](#) for helpful articles to resolve common technical issues. The chatbot and support agents are also available to assist via the mobile or desktop versions.

Translation and accessibility issues: Make sure your device's language settings are updated. In courses that use auto-translation, some content may not be fully translated, such as downloadable files, attachments, etc. You can refer to the guide on [translation for course content](#) for more support.

Certificates and financial aid: If you are unable to pay for the course, you can read this article to [apply for financial aid](#) which explains how to submit an application. If you have already applied and are experiencing issues, read the article on [solving problems with financial help](#) which provides guidance on common problems and how to resolve them. Check out how to [get a course certificate](#) article if you want to learn more on the requirements for earning a Course Certificate, including how to download or share it once completed.

Course resources and completion issues: Coursera tracks your progress differently depending on the type of activity. Videos are marked as complete automatically once you've watched 90% of the content. For readings and embedded activities, you'll need to manually click 'Mark as complete' once finished unless you're using a smartphone where you need to click 'Next' instead. Quizzes are marked complete when all questions are answered and submitted, even if the answers are incorrect as these are designed to help you check your understanding and decide whether a review is needed. Discussion forums are marked complete once you contribute a post, and we encourage thoughtful participation, as engaging with diverse perspectives can enhance your learning. If you encounter any issues with a resource completion, click the flag icon to report it (see below more information).

General issues: If you find a problem with a course while using Coursera, we want to know about it. If something doesn't work as expected, use this article to learn how to [report a problem with the course](#).

Disclaimer: The learning materials on this course, including video lectures, have been taken from the Computational Mathematics module on the University of London's BSc Computer Science programme.

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